

Finite Free Stam Inequality

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1 Problem statement

4. Let $p(x)$ and $q(x)$ be two monic polynomials of degree n :

$$p(x) = \sum_{k=0}^n a_k x^{n-k} \quad \text{and} \quad q(x) = \sum_{k=0}^n b_k x^{n-k}$$

where $a_0 = b_0 = 1$. Define $p \boxplus_n q(x)$ to be the polynomial

$$(p \boxplus_n q)(x) = \sum_{k=0}^n c_k x^{n-k}$$

where the coefficients c_k are given by the formula:

$$c_k = \sum_{i+j=k} \frac{(n-i)!(n-j)!}{n!(n-k)!} a_i b_j$$

for $k = 0, 1, \dots, n$. For a monic polynomial $p(x) = \prod_{i \leq n} (x - \lambda_i)$, define

$$\Phi_n(p) := \sum_{i \leq n} \left(\sum_{j \neq i} \frac{1}{\lambda_i - \lambda_j} \right)^2$$

and $\Phi_n(p) := \infty$ if p has a multiple root. Is it true that if $p(x)$ and $q(x)$ are monic real-rooted polynomials of degree n , then

$$\frac{1}{\Phi_n(p \boxplus_n q)} \geq \frac{1}{\Phi_n(p)} + \frac{1}{\Phi_n(q)}?$$

2 Preliminaries

2.1 An equivalent formula for Φ_n via derivatives

Let p be monic with distinct real roots $\lambda_1, \dots, \lambda_n$. Using

$$\frac{p'(x)}{p(x)} = \sum_{j=1}^n \frac{1}{x - \lambda_j},$$

one checks (e.g. by factoring $p(x) = (x - \lambda_i)p_i(x)$) that

$$\sum_{j \neq i} \frac{1}{\lambda_i - \lambda_j} = \frac{p''(\lambda_i)}{2p'(\lambda_i)}. \quad (1)$$

Hence

$$\Phi_n(p) = \frac{1}{4} \sum_{i=1}^n \left(\frac{p''(\lambda_i)}{p'(\lambda_i)} \right)^2, \quad (\text{distinct roots}). \quad (2)$$

If p has a multiple root, $\Phi_n(p) = \infty$ by definition and the desired inequality is automatically true.

2.2 Affine covariance

Two basic invariances are immediate from the root-difference definition:

- **Translation:** If $p_t(x) = p(x - t)$ then $\Phi_n(p_t) = \Phi_n(p)$.
- **Scaling:** If $p^{(s)}(x) = s^{-n}p(sx)$ (so $p^{(s)}$ is monic and has roots λ_i/s), then

$$\Phi_n(p^{(s)}) = s^2 \Phi_n(p), \quad \text{so} \quad \frac{1}{\Phi_n(p^{(s)})} = \frac{1}{s^2} \cdot \frac{1}{\Phi_n(p)}. \quad (3)$$

Moreover, the definition of $\boxplus_n$ shows it is compatible with translations and scalings (coefficients transform in the expected way). Consequently, one may freely impose an affine normalization (e.g. “center” the roots) when proving the inequality.

3 Mechanism explored, accepted, rejected)

This section records the principal *mechanisms* (proof strategies/lemmas) explored.

3.1 Accepted mechanisms (used in the proof)

- A1. Affine reduction.** Use translation/scaling covariance to normalize polynomials (especially for $n = 3$) without loss of generality.
- A2. Derivative representation of Φ_n .** Use (2) to rewrite Φ_n in algebraic terms.
- A3. Gap parametrization for $n = 3$.** For ordered roots $\lambda_1 < \lambda_2 < \lambda_3$, set gaps $a = \lambda_2 - \lambda_1$, $b = \lambda_3 - \lambda_2$ and compute Φ_3 explicitly.
- A4. Additive invariants for centered cubics.** After centering (zero quadratic term), two invariants (A, B) add under $\boxplus_3$.
- A5. Convexity/Jensen step.** The $n = 3$ inequality reduces to Jensen for $x \mapsto x^2$ on a weighted average of ratios.

3.2 Rejected mechanisms (with explicit obstruction)

- R1. Fractional $\boxplus_3$ -powers preserve real-rootedness.** A natural semigroup idea is: define $p^{\boxplus_3 t}$ by scaling *finite-free cumulants* by t and try to prove the inequality by interpolation (analogous to heat-flow proofs of Stam's inequality). This fails: real-rootedness is not preserved for $t \in (0, 1)$ in general. A fully worked counterexample is given in Appendix A.
- R2. Binary invariant-theory ansatz for all $n = 4$.** An attempt to mimic the $n = 3$ proof by expressing $1/\Phi_4$ in terms of two additive invariants (as for the classical cubic discriminant identity) runs into a structural obstacle: the simplest low-degree invariants for quartics are *not* additive under $\boxplus_4$ in any straightforward normalization, so the Jensen reduction does not close.

3.3 Partially explored (progress, but not closed)

- P1. Even quartics.** For $p(x) = (x^2 - a^2)(x^2 - b^2)$ one can compute $\Phi_4(p)$ in closed form (Section 6). Under $\boxplus_4$, even quartics form a closed subfamily with a simple coefficient update rule. This reduces Problem 4 on that subfamily to an explicit two-parameter inequality, but we do not claim a complete proof here.

4 Degree $n = 2$: the inequality holds (with equality)

Theorem 4.1. *Problem 4 holds for $n = 2$.*

Proof. Let $p(x) = x^2 + a_1x + a_2$ and $q(x) = x^2 + b_1x + b_2$ be monic with real roots. From the definition of $\boxplus_2$,

$$c_1 = a_1 + b_1, \quad c_2 = a_2 + b_2 + \frac{1}{2}a_1b_1,$$

so $(p \boxplus_2 q)(x) = x^2 + c_1x + c_2$.

By translation invariance we may center each polynomial (eliminate the linear term): for a centered monic quadratic $r(x) = x^2 + u$ with $u \leq 0$, the roots are $\pm\sqrt{-u}$ and hence

$$\Phi_2(r) = \left(\frac{1}{2\sqrt{-u}}\right)^2 + \left(\frac{1}{-2\sqrt{-u}}\right)^2 = \frac{1}{2(-u)}.$$

Thus $1/\Phi_2(r) = 2(-u)$.

In centered coordinates, $\boxplus_2$ simply adds the constant terms, so u adds. Therefore

$$\frac{1}{\Phi_2(p \boxplus_2 q)} = \frac{1}{\Phi_2(p)} + \frac{1}{\Phi_2(q)}.$$

This implies the claimed inequality for $n = 2$ (indeed, equality). □

5 Degree $n = 3$: complete proof

5.1 Computing Φ_3 in gap coordinates

Let p be monic with *distinct* real roots $\lambda_1 < \lambda_2 < \lambda_3$. Define gaps

$$a = \lambda_2 - \lambda_1 > 0, \quad b = \lambda_3 - \lambda_2 > 0.$$

Lemma 5.1. *With a, b as above,*

$$\Phi_3(p) = 2 \left(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{(a+b)^2} \right).$$

Proof. For $i = 1, 2, 3$ define $S_i = \sum_{j \neq i} \frac{1}{\lambda_i - \lambda_j}$. A direct computation yields

$$S_1 = -\frac{1}{a} - \frac{1}{a+b}, \quad S_2 = \frac{1}{a} - \frac{1}{b}, \quad S_3 = \frac{1}{b} + \frac{1}{a+b}.$$

Expanding $S_1^2 + S_2^2 + S_3^2$ shows that all mixed terms cancel, leaving the stated formula. $\square$

5.2 Centered cubics and additive invariants

By translation invariance, we may center p (replace x by $x+t$) so that the x^2 coefficient vanishes. Thus we may assume

$$p(x) = x^3 + ux + v. \quad (4)$$

For a centered cubic with gaps a, b as above (after centering), one can express u, v in terms of a, b ; it is convenient to package them into:

$$A := a^2 + ab + b^2, \quad B := (a-b)(2a^2 + 5ab + 2b^2). \quad (5)$$

Lemma 5.2. *For a centered cubic $p(x) = x^3 + ux + v$ with ordered real roots and gaps a, b , the quantities*

$$A = -3u, \quad B = 27v$$

agree with (5).

Proof. Let the roots be $\lambda_1, \lambda_2, \lambda_3$ with gaps a, b and with $\lambda_1 + \lambda_2 + \lambda_3 = 0$. A standard parametrization is

$$\lambda_1 = -\frac{2a+b}{3}, \quad \lambda_2 = \frac{a-b}{3}, \quad \lambda_3 = \frac{a+2b}{3}.$$

Then $u = \lambda_1\lambda_2 + \lambda_1\lambda_3 + \lambda_2\lambda_3 = -\frac{1}{3}(a^2 + ab + b^2)$ and $v = -(\lambda_1\lambda_2\lambda_3) = \frac{1}{27}(a-b)(2a^2 + 5ab + 2b^2)$, giving the claim. $\square$

Lemma 5.3. *With A, B as above, the discriminant $\Delta(p) = \prod_{i < j} (\lambda_i - \lambda_j)^2$ satisfies*

$$\Delta(p) = a^2b^2(a+b)^2 = \frac{4A^3 - B^2}{27}.$$

Proof. The equality $\Delta = a^2b^2(a+b)^2$ is immediate from gap coordinates. For the identity $4A^3 - B^2 = 27a^2b^2(a+b)^2$, expand both sides as polynomials in a, b (one checks it on the nose; it is the classical binary cubic invariant relation specialized to these parameters). $\square$

5.3 Closed form for $1/\Phi_3$

Proposition 5.4. *For a monic centered cubic $p(x) = x^3 + ux + v$ with distinct real roots, define*

$$A := -3u > 0, \quad B := 27v.$$

Then

$$\frac{1}{\Phi_3(p)} = \frac{\Delta(p)}{2A^2} = \frac{4A^3 - B^2}{54A^2} = \frac{2A}{27} - \frac{1}{54} \left(\frac{B}{A} \right)^2. \quad (6)$$

Proof. By Lemma 5.1, $\Phi_3 = 2(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{(a+b)^2})$, hence

$$\frac{1}{\Phi_3} = \frac{1}{2} \cdot \frac{1}{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{(a+b)^2}} = \frac{a^2 b^2 (a+b)^2}{2(a^2 + ab + b^2)^2} = \frac{\Delta}{2A^2}.$$

Using Lemma 5.3 gives $\Delta = (4A^3 - B^2)/27$, yielding (6). $\square$

5.4 Additivity of (A, B) under $\boxplus_3$ in centered form

Let $p(x) = x^3 + u_p x + v_p$ and $q(x) = x^3 + u_q x + v_q$ be centered monic cubics. From the definition of $\boxplus_3$ (explicit coefficient computation),

$$(p \boxplus_3 q)(x) = x^3 + (u_p + u_q)x + (v_p + v_q).$$

Thus u and v add, and therefore

$$A(p \boxplus_3 q) = A(p) + A(q), \quad B(p \boxplus_3 q) = B(p) + B(q). \quad (7)$$

5.5 Stam inequality for $n = 3$

Theorem 5.5. *Problem 4 holds for $n = 3$.*

Proof. If p or q has a multiple root then $\Phi_3 = \infty$ and the inequality is trivial, so assume distinct roots.

By translation invariance, center both polynomials: replace p, q by translates so that their x^2 coefficients vanish. This does not change Φ_3 and is compatible with $\boxplus_3$ (centering each yields a centered convolution).

Let A_p, B_p and A_q, B_q be defined as in Proposition 5.4. Then by (7),

$$A_{p \boxplus_3 q} = A_p + A_q, \quad B_{p \boxplus_3 q} = B_p + B_q.$$

Using (6),

$$\frac{1}{\Phi_3(p \boxplus_3 q)} - \frac{1}{\Phi_3(p)} - \frac{1}{\Phi_3(q)} = -\frac{1}{54} \left[\left(\frac{B_p + B_q}{A_p + A_q} \right)^2 - \left(\frac{B_p}{A_p} \right)^2 - \left(\frac{B_q}{A_q} \right)^2 \right].$$

Set weights $\alpha = \frac{A_p}{A_p + A_q}$ and $\beta = \frac{A_q}{A_p + A_q}$ (so $\alpha, \beta \geq 0$ and $\alpha + \beta = 1$) and define $x_p = \frac{B_p}{A_p}$, $x_q = \frac{B_q}{A_q}$. Then

$$\frac{B_p + B_q}{A_p + A_q} = \alpha x_p + \beta x_q.$$

Since $x \mapsto x^2$ is convex,

$$(\alpha x_p + \beta x_q)^2 \leq \alpha x_p^2 + \beta x_q^2,$$

which is equivalent to

$$\left(\frac{B_p + B_q}{A_p + A_q} \right)^2 \leq \frac{A_p}{A_p + A_q} \left(\frac{B_p}{A_p} \right)^2 + \frac{A_q}{A_p + A_q} \left(\frac{B_q}{A_q} \right)^2.$$

Multiplying by $(A_p + A_q)$ yields precisely

$$\left(\frac{B_p + B_q}{A_p + A_q} \right)^2 \leq \left(\frac{B_p}{A_p} \right)^2 + \left(\frac{B_q}{A_q} \right)^2.$$

Hence the bracketed term above is ≤ 0 , so the whole difference is ≥ 0 and therefore

$$\frac{1}{\Phi_3(p \boxplus_3 q)} \geq \frac{1}{\Phi_3(p)} + \frac{1}{\Phi_3(q)}.$$

□

Remark 5.6 (Equality for $n = 3$). The Jensen step is tight exactly when $x_p = x_q$, i.e. $\frac{B_p}{A_p} = \frac{B_q}{A_q}$. In gap language this means the two centered cubics have the same dimensionless shape parameter (equivalently, the same ratio $v/(-u)^{3/2}$), i.e. they are affine rescalings of one another.

6 Partial progress for $n = 4$: even quartics

This section records a concrete computation and a reduction for an important subfamily.

6.1 Closed form for Φ_4 on $p(x) = (x^2 - a^2)(x^2 - b^2)$

Let $0 < a < b$ and

$$p(x) = (x^2 - a^2)(x^2 - b^2) = x^4 - (a^2 + b^2)x^2 + a^2b^2,$$

with roots $-b, -a, a, b$.

Lemma 6.1. For $p(x) = (x^2 - a^2)(x^2 - b^2)$ with $0 < a < b$,

$$\Phi_4(p) = \frac{4}{(b-a)^2} + \frac{4}{(a+b)^2} + \frac{1}{2a^2} + \frac{1}{2b^2}.$$

Proof. Write the roots $\lambda_1 = -b, \lambda_2 = -a, \lambda_3 = a, \lambda_4 = b$. The inner sums $S_i = \sum_{j \neq i} \frac{1}{\lambda_i - \lambda_j}$ satisfy $S_4 = -S_1$ and $S_3 = -S_2$ by symmetry, hence $\Phi_4(p) = 2(S_1^2 + S_2^2)$. A direct expansion shows all mixed terms cancel, leaving

$$S_1^2 + S_2^2 = \frac{2}{(b-a)^2} + \frac{2}{(a+b)^2} + \frac{1}{4a^2} + \frac{1}{4b^2},$$

and multiplying by 2 yields the claim. □

6.2 Closure of even quartics under $\boxplus_4$ and an explicit update rule

Write a general even monic quartic as

$$p(x) = x^4 + u_p x^2 + v_p, \quad q(x) = x^4 + u_q x^2 + v_q.$$

A coefficient computation from the definition of $\boxplus_4$ gives:

$$(p \boxplus_4 q)(x) = x^4 + (u_p + u_q)x^2 + \left(v_p + v_q + \frac{1}{6}u_p u_q\right). \quad (8)$$

Thus even quartics are closed under $\boxplus_4$ and the update is explicit.

Remark 6.2 (Status). Combining Lemma 6.1 (which gives Φ_4 on the two-parameter real-rooted locus) with (8) reduces Problem 4 on this subfamily to an explicit inequality in two real parameters (after affine normalization). This reduction is *validated*; we do *not* claim the resulting inequality has been fully closed here.

A Counterexample: fractional $\boxplus_3$ -powers can destroy real-rootedness

We record the promised obstruction to the “fractional $\boxplus_3$ -power” interpolation mechanism.

Step 1: additive coordinates for $\boxplus_3$

For a monic cubic $p(x) = x^3 + a_1x^2 + a_2x + a_3$, define

$$\kappa_1(p) := -\frac{a_1}{3}, \quad \kappa_2(p) := a_2 - \frac{a_1^2}{3}, \quad \kappa_3(p) := a_3 - \frac{a_1a_2}{3} + \frac{2a_1^3}{27}.$$

A direct computation from the coefficient formula for $\boxplus_3$ shows that

$$\kappa_j(p \boxplus_3 q) = \kappa_j(p) + \kappa_j(q) \quad (j = 1, 2, 3).$$

These are the natural additive coordinates (finite-free cumulants in degree 3).

Step 2: define a fractional power and exhibit failure of real-rootedness

Define $p^{\boxplus_3 t}$ by scaling these cumulants:

$$\kappa_j(p^{\boxplus_3 t}) = t \kappa_j(p), \quad j = 1, 2, 3,$$

and reconstructing the coefficients (a_1, a_2, a_3) from $(\kappa_1, \kappa_2, \kappa_3)$.

Consider

$$p(x) = x^3 - 11x^2 + 10x,$$

which is real-rooted: $p(x) = x(x-1)(x-10)$.

Compute cumulants:

$$\kappa_1(p) = \frac{11}{3}, \quad \kappa_2(p) = 10 - \frac{121}{3} = -\frac{91}{3}, \quad \kappa_3(p) = 0 - \frac{(-11) \cdot 10}{3} + \frac{2(-11)^3}{27} = -\frac{1672}{27}.$$

For $t = \frac{1}{2}$,

$$\kappa_1 = \frac{11}{6}, \quad \kappa_2 = -\frac{91}{6}, \quad \kappa_3 = -\frac{836}{27}.$$

Reconstruct coefficients via

$$a_1 = -3\kappa_1, \quad a_2 = \kappa_2 + \frac{a_1^2}{3}, \quad a_3 = \kappa_3 + \frac{a_1a_2}{3} - \frac{2a_1^3}{27},$$

giving

$$p^{\boxplus_3 1/2}(x) = x^3 - \frac{11}{2}x^2 - \frac{61}{12}x - \frac{671}{72}.$$

Its discriminant is

$$\Delta = -\frac{644221}{54} < 0,$$

so it has only one real root. Therefore, the fractional $\boxplus_3$ -power operation does *not* preserve real-rootedness in general, and semigroup/heat-flow arguments based on fractional powers cannot be used without additional structure.